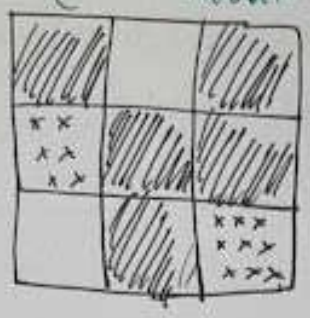
Image Histogram:

(NOTE: Mean of the histogram tells me about brightness; variance tells me about contrast!

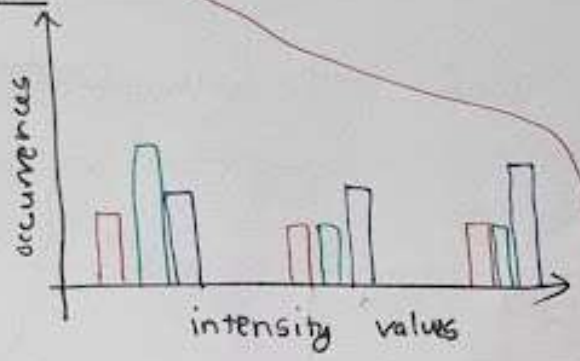
Decrease mean → increase brightness

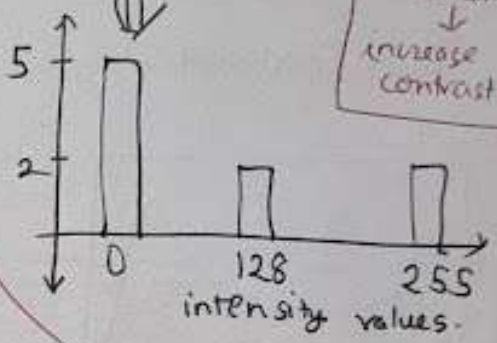
Increase variance → increase contrast



0	255	0
128	0	0
255	0	128

For RGB image:





> Histograms provide us with a lot of useful information about the image which can be used to:

a) analyze the image
b) enhance the image
c) change/transform the image

} Done through transformation function that maps intensity values into new intensity values

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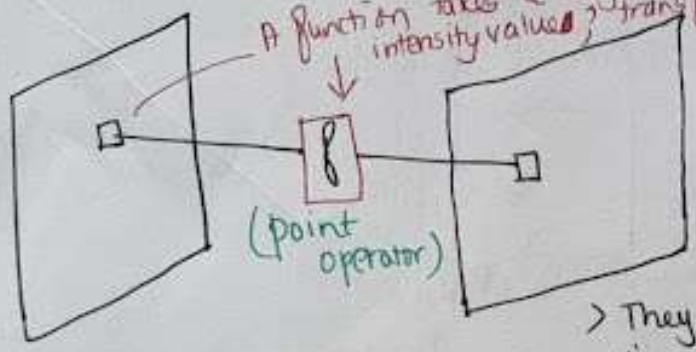
> Used in
- → photography
- → photogrammetry
- → image processing
- → computer vision

} To enforce certain property in your image!

How are those Histograms manipulated?



> This is an individual decision for every intensity value; so it's basically a lookup table!

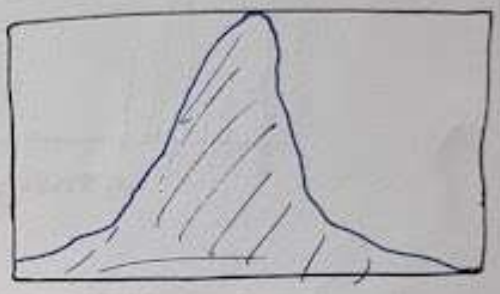
> They can be expressed often via a curve known in photography as a tone curve.

Mathematically, the functions are point operators used to change contrast, brightness, and other properties.

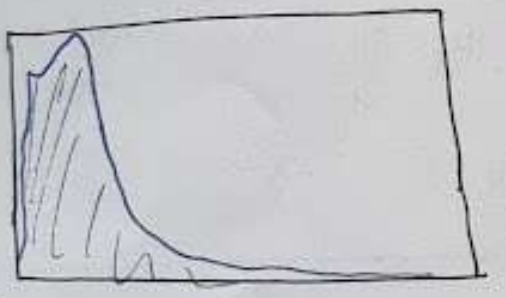
HISTOGRAMS:

High Contrast:

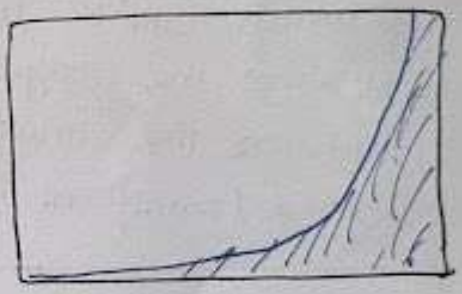
Low Contrast:



Under exposed:



Overexposed:



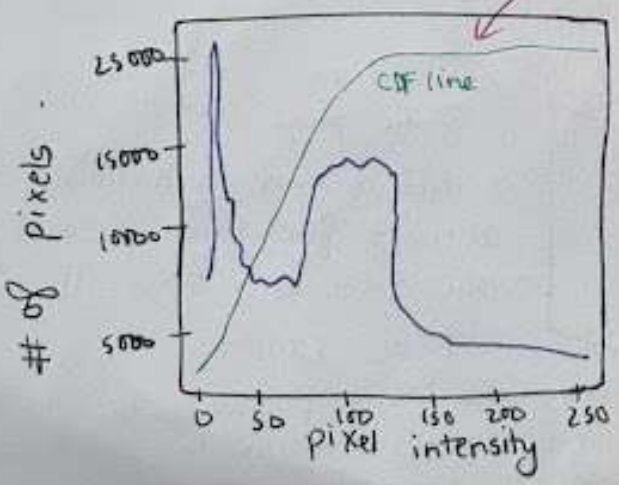
Histogram Equalization:

CDF = Cumulative Distribution Function.
(cumulative sum of all the pixels)

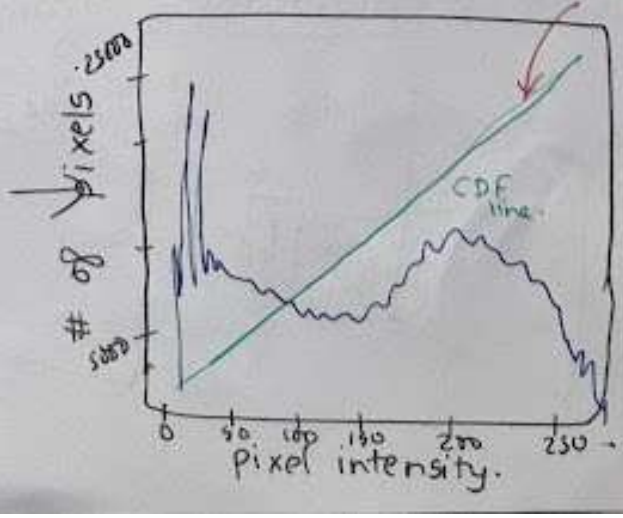
> What is histogram equalization?

→ A method of redistributing pixel intensities so that all intensities are equally distributed.

CDF ramping up quickly & leveling off.



CDF shows linear ramp. means more or less equally distributed pixel values



> How does histogram equalization work?

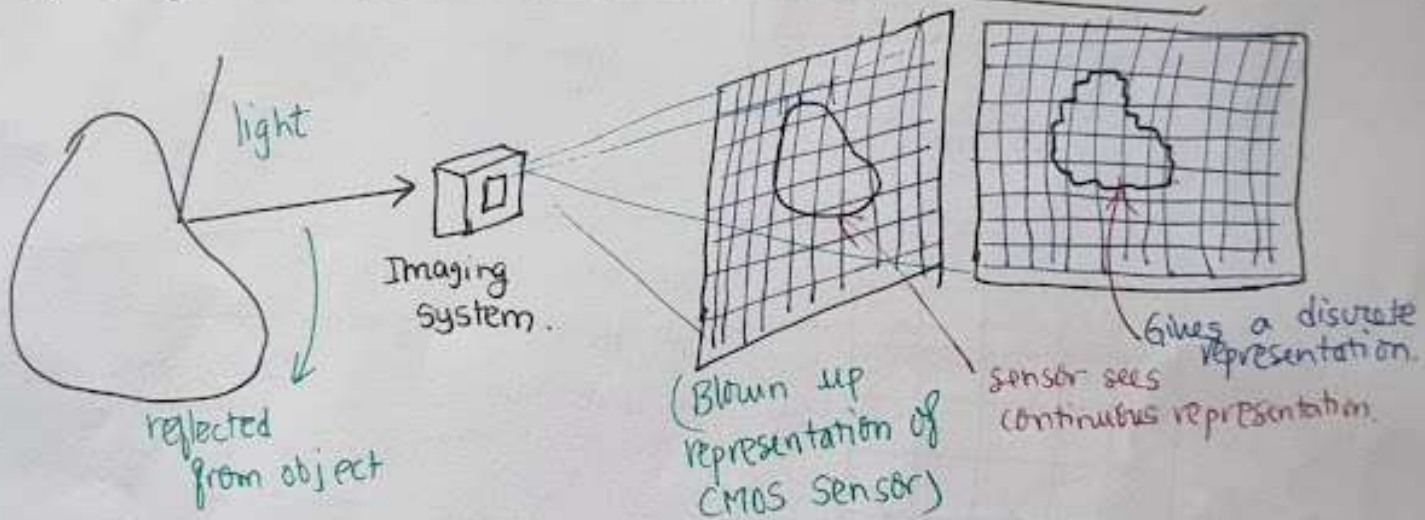
1) Compute CDF (cumulative distribution function)
2) Map input intensities to new intensities based on CDF.
3) Create new image from mapping.

CLAHE → Contrast Limiting Adaptive Equalization.

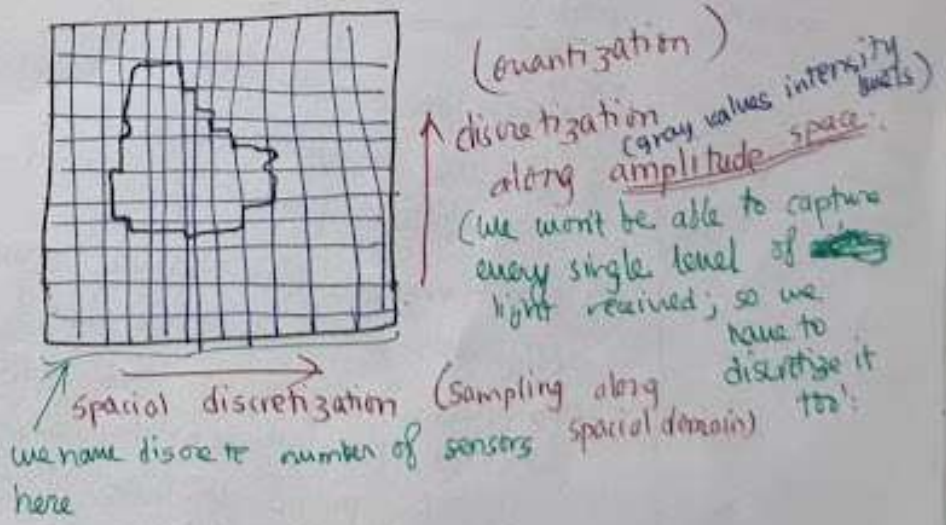
Usage: For enhancing local contrast!

(Illumination)

DIGITAL IMAGE FUNDAMENTALS



> The discretization happens in two different directions:



See Image in Snippets Folder

pixel: picture element.

256 X 256 pixels } Spacial samples.
8-bit } amplitude samples (quantized levels or depths!)

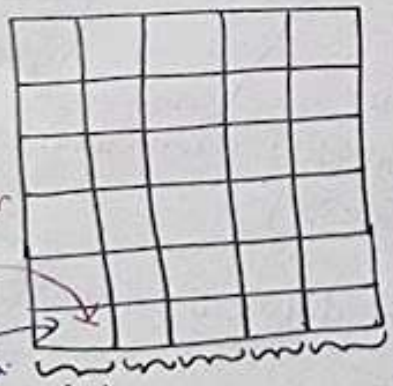
Expensive Cameras Capture : Each 3 full images.
1) Red
2) Green
3) Blue

Cheaper cameras : Do sth called mosiaqe; interleaves the colors when capturing:

R	G	R	G	R
B	R	B	R	B
R	G	R	G	R
B	R	B	R	B
R	G	R	G	R
B	R	B	R	B

mosiaque.

→ Here we have B, we need G & R; (which will be obtained by interpolating from the neighbors!)



Pixels are stored From "Left to Right" and "Top to Bottom".

multiple color in contents of area

average color of the pixel area will be used.

pixel size.

depends on quantization levels: more levels more clean the pic.

⇒ Called bits-per-pixel levels are

⇒ a black & white image has 8-bits per pixel.

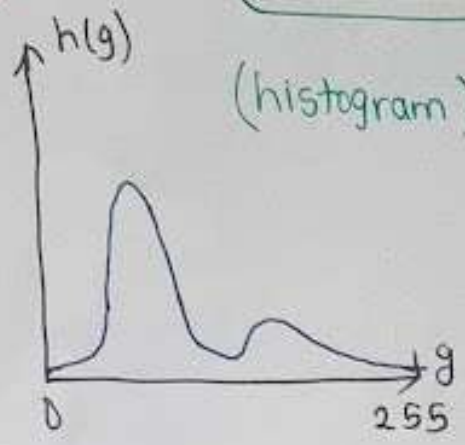
⇒ For images, the sample rate is measured in dots per inch (i.e. dpi)

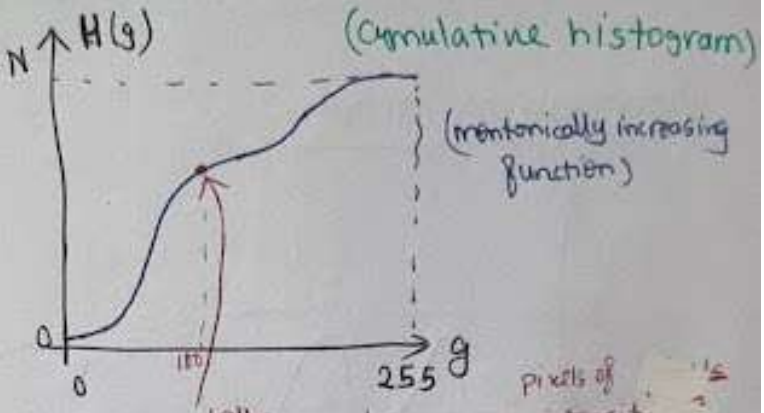
⇒ An image digitized at 100 dpi will have 100 pixels per inch of a photograph. In 2D; its 100 X 100 pixels i.e. 10,000 pixels per inch.

Cumulative Histogram:

Definition:

$$\boxed{H(g) = \sum_{x=0}^{g} h(x) = \#\text{pixel with } g(i,j) \le g}$$



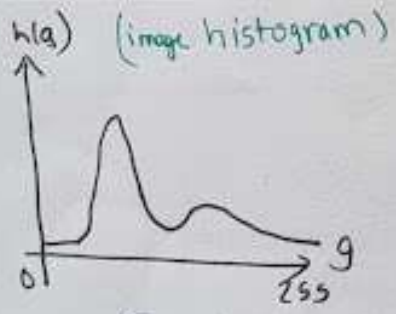


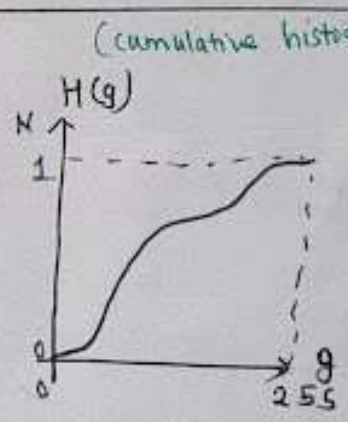
tells me how many pixels of intensity that value or smaller.
(all pixel with intensity 100 or less)

i.e

$$\begin{cases} H(0) = h(0) \\ H(1) = h(0) + h(1) \\ H(5) = h(0) + h(1) + h(2) \cdots + h(5) \end{cases}$$

Empirical PDF & CDF





(This f() is Integral over our pdf)

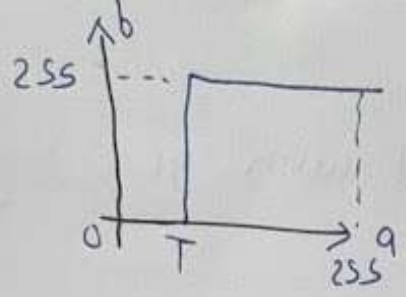
Empirical pdf. $\left[p(g) = \dfrac{h(g)}{N} \right]$

Empirical CDF $\left[F(g) = P(g(i,j) \le g) = \dfrac{H(g)}{N} \right]$

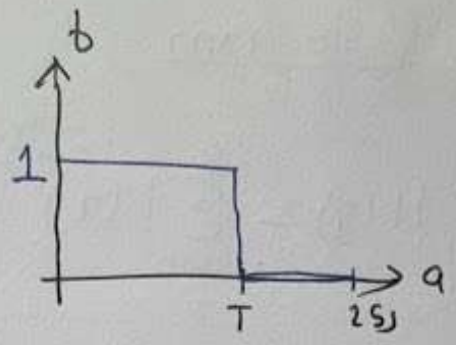
NOTE: Image histogram tells us something about the image; i.e. what we actually see in the image ("not what is depicted there" but how it $\underbrace{\text{looks like}}_{\text{i.e. brightness or contrast}}$".)

Question: Can two different pictures have the same image histogram?

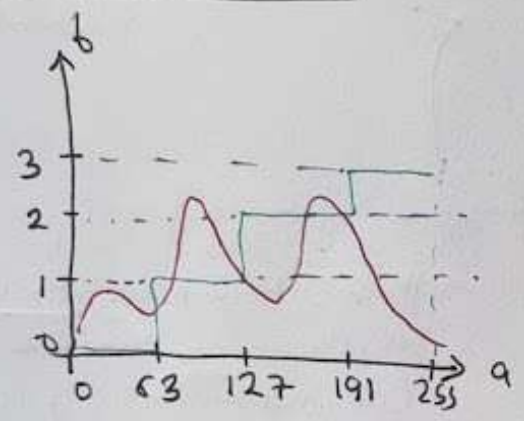
Binary Image Graph:



OR



Quantization Graph:



Lookup tables (tone curves):